Jenna A. Lee

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Dept. of Geosciences
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PROFILE SUMMARY:

Recent Ph.D. recipient with substantial research, teaching, and outreach involvement. Committed to creating an engaging learning environment by utilizing active learning strategies and tailoring teaching methods to individuals with equity and inclusion in mind. Excited to learn from and collaborate with mentors, peers, and students alike.

EDUCATION:

Ph.D. - Princeton University Dept. of Geosciences
2019 - 2025
Graduate Advisor: Bess B. Ward

M.A. - Princeton University Dept. of Geosciences
2019 - 2021
Graduate Advisor: Bess B. Ward

B.Sc. (Honors) - UC Irvine Dept. of Earth System Science
2014 - 2018
Dual Specialization: Oceanography and Hydrology & Terrestrial Ecosystems
Minor: Global Sustainability
Undergraduate Advisors: Adam C. Martiny & François Primeau

RESEARCH & WORK EXPERIENCE:

Lab Specialist **(2026- present)**
Dept. of Geosciences, Princeton University
- Executed and optimized experimental protocols for seawater nutrient measurements in a new lab setting, including equipment setup, troubleshooting, and development of updated SOPs and safety procedures.

Graduate Student Researcher
Dept. of Geosciences, Princeton University **(2019-2025)**
Supervised by Dr. Bess B. Ward
- Designed and conducted field mesocosm experiments investigating the temporal variability in eukaryotic community composition and biogeochemical processes such as primary productivity rates
- Extracted, prepared, and analyzed 18S amplicon sequences to investigate community composition, diversity, and potential eukaryotic interactions via co-occurrence network construction (SPIEC-EASI)
- Generated bacterial metagenomically assembled genomes (MAGs) from raw metagenomic sequences and investigated potential bacterial-eukaryotic interactions using cross-domain network construction
- Evaluated nitrogen transport, CO_2 fixation, and specific uptake rates through the analysis of ^{13}C- and ^{15}N-labeled incubations and ambient nutrients
- Assisted in organizing multiple field expeditions to Chesapeake Bay (2020-2022) and the Eastern Tropical South Pacific (2023) and participated in related field experiments and data collection

RESEARCH & WORK EXPERIENCE (CONTINUED):

Junior Lab Specialist
Adam Martiny Lab, Dept. of Earth System Science, UC Irvine (2018-2019)

- Participated in AMT28 BAS research expedition (2018) for sample collection
- Performed sample processing, preliminary data analysis, and modeled the diel variability of particulate organic matter, specifically carbon, nitrogen, and phosphorus
- Performed DNA extraction and preparation of samples for metagenomic sequencing

Undergraduate Researcher
Dept. of Earth System Science and Dept. of Ecology and Evolutionary Biology, UC Irvine (2017-2018)
Supervised by Dr. Adam Martiny

- Determined the relationship between latitudinal variations in phytoplankton stoichiometry and environmental factors in the Pacific Ocean to assist in understanding global variations from the Redfield Ratio, and used this information combined with physical oceanographic parameters to model ecologically distinct marine biomes
- Participated in IO7N NOAA research expedition (2018) for sample collection
- Optimized the particulate organic phosphorus assay protocol and performed analysis of POM

Undergraduate Researcher
Dept. of Earth System Science, UC Irvine (2016-2017)
Supervised by Dr. François Primeau

- Modeled the relationship between maximum entropy and high biological productivity regions of the ocean using the Stommel ('61) model for thermohaline circulation and multiple equilibria

Chemistry Laboratory Stockroom Assistant
Dept. of Chemistry, UC Irvine (2016-2018)

- Organized, prepared, and maintained chemical reagents and equipment for 2-3 concurrent upper division teaching labs for biological, organic, and analytical chemistry
- Addressed Teaching Assistants' and undergraduate students' questions, concerns, and requests regarding lab materials
- Assessed and assisted in the handling of emergency situations such as evacuations, chemical spills, and minor laboratory injuries
- Cleaned laboratory equipment, maintained safe and organized lab spaces, and managed the handling of hazardous waste

SCIENTIFIC CRUISES & FIELD WORK:

RR2311, University-National Oceanographic Laboratory System (UNOLS) (2023)
5-week cruise (Nov-Dec) aboard the R/V Roger Revelle; Chief Scientist: Bess B. Ward (Princeton)

- Organized the collection, storage, and onboard analysis of in-situ nutrients
- Worked in an 8-person research team to plan, manage, and perform several concurrent experiments measuring N_2O cycling process rates in the oxygen minimum zone off the coast of Peru

CB20, CB21, CB22, UNOLS (2020, 2021, 2022)
Several 7-10 day cruises aboard the R/V Hugh Sharp; Chief Scientist: Amal Jayakumar (Princeton)

- Organized and executed week-long bloom mesocosm experiments, sampling for DNA, particulate matter, photopigments, ambient nutrients, and nutrient uptake rates multiple times per day
- Assisted in planning, managing, and participating in experiments to measure N_2O concentrations and cycling process rates in the seasonally anoxic Chesapeake Bay

AMT28, British Antarctic Survey (BAS) (2018)
5-week cruise aboard the RSS James Clark Ross; Chief Sci. Glen Tarran (Plymouth Marine Laboratory)

- Co-led the 24-hr sampling of particulate matter, eDNA, and protein samples from the U.K. to the Falkland Islands

SCIENTIFIC CRUISES & FIELD WORK (CONTINUED):

IO7N, National Oceanic and Atmospheric Administration (NOAA) **(2018)**
7-week cruise aboard the R/V Ronald Brown; Chief Scientist: Denis Volkov (NOAA)
- Co-led the 24-hr sampling of POM and eDNA from South Africa to India

TEACHING & MENTORING:

Community College Teaching Fellow (CCTF)
Program for Community College Engagement, Princeton University
Essentials of Oceanography (SCI-160) Middlesex College Dept. of Natural Sciences
Spring 2025
- Guest lectured on several topics including seawater density, surface ocean circulation, coastal processes, and an introduction to biological oceanography. Lectures highlighted ocean processes in the context of the global Earth System with a focus on ocean-atmosphere interactions.
- Developed new course materials such as lecture content and slides, lab exercises, in-class demonstrations, and exam questions. Assisted in updating previous course materials.
- Led students in the creation of scientific research posters and provided feedback as part of a final project

Teaching Assistant (TA)
Ocean, Atmosphere, and Climate (GEO202) Princeton University Dept. of Geosciences
Spring 2024
- Coordinated with faculty, course manager, and fellow TA members to synthesize and address student feedback for a 50+ student course throughout the semester
- Led 1-hour “mini-lab” precepts for 3 sections of up to 10 students each week
- Assisted in the development of course materials such as precept handouts, problem sets, and exams
- Graded and provided feedback on students’ precept assignments, exams, and homework assignments in a timely manner
- Hosted weekly office hours and additional supplemental meetings for undergraduate students

Climate: Past, Present, and Future (GEO102) Princeton University Dept. of Geosciences
Fall 2021
- Coordinated with faculty, course manager, and 4 other TAs to ensure content was presented consistently across labs and lecture for a 150+ student course
- Guided 10-15 students through weekly 3-hour labs using hands-on demonstrations of course concepts and developing skills in experimental design, data collection/analysis, and scientific communication
- Synthesized and communicated student feedback to faculty and course manager, addressing student concerns and more effectively achieving faculty goals
- Assisted in the development of lab materials, problem sets, and exams
- Graded and provided feedback on labs, exams, and assignments and hosted office hours as above

Fall 2020
- Collaborated with faculty and course managers in the transition to online learning by reviewing and modifying new online materials (e.g. exams and quizzes), primarily implemented through Canvas, for a 150+ student course
- Hosted weekly Zoom-based office hours in addition to the above grading responsibilities

Mentorship
Princeton Women in Geosciences (PWiGs) – Peer Mentor **(2022-present)**
- Mentored junior graduate students in the Geosciences dept. in navigating academia
- Assisted in goal-setting, time-management, general research skills, and conflict resolution

TEACHING & MENTORING (CONTINUED):

Mentorship

Princeton University High Meadows Environmental Institute (HMEI) – Undergraduate Mentor

Summer 2025 – Molly Thacker, Ottilie Sykes, & Darren McKeogh

- Developed and oversaw a 2-week intensive training for 3 undergraduate students in preparation for a research cruise. Training included colorimetric nutrient analysis (spectrophotometry), anoxic seawater incubations, and general lab and field work techniques.

Princeton University High Meadows Environmental Institute (HMEI) – Undergraduate Mentor

Summer 2024 – Oleksandr (Sasha) Mykhantso & Celia Murphy-Braunstein

- Trained undergraduate students on lab techniques for water quality testing, oversaw their lab work, and reviewed and provided feedback on research presentations.

Summer 2021 – Adira Smirnov

- Developed a bioinformatics research project for an undergraduate student's 10-week internship
- Provided 1-on-1 instruction of coding basics, data acquisition, and statistical analysis.
- Assisted in the student-led development of a research presentation

OUTREACH & SERVICE:

Outreach

Foundation Year Charter School - Teaching Demonstration **(February 2024)**

- Instructed K-12 students on applications of the scientific method using hands-on activities and tailored to multiple classes at different grade levels
- Led students in activities demonstrating respiration and measuring CO_2 production by yeast, including skills in hypothesis formulation and graph interpretation
- Introduced students to microscopy and led discussions of morphology using phytoplankton slides

Exhibitor at Spring into Science **(2023, 2024, 2025)**

- Presented to 4th-10th graders and their families on the topics of coral reefs and climate change to help foster an interest in science and scientific careers
- Led interactive demonstrations to communicate basic chemistry and physics concepts
- Relevant articles: [2023], [2024], [2025]

Exhibitor at the Young Women's' Conference (PPPL) **(March 2023)**

- A conference of 400+ 7th-10th grade girls with the goal of introducing young women to the variety of science and engineering careers and facilitating networking
- Led interactive demonstrations on ocean acidification and cloud formation

Service Work

Departmental Diversity Committee **(2022-2025)**

- Served as a communications liaison between faculty, post-docs, graduate students, and undergraduates on issues related to DEIA and organized monthly meetings with other Diversity Committee members
- Served on an additional non-faculty committee which managed the $20,000/year Diversity, Equity, Inclusion, and Accessibility (DEIA) Fund. Reviewed proposals for the DEIA Fund, assisted in transferring funds, and followed up with successful proposals to assess outcomes
- Assisted in organizing diversity and inclusion events within and beyond the department, including several outreach events and a mental health first aid training course
- Created and implemented the "Field Code of Conduct Follow-up Form" and follow-up procedure for all University-sponsored travel. Reviewed all follow-up forms and summarized all responses and recommendations for trip organizers to improve future travel arrangements

OUTREACH & SERVICE (CONTINUED):

Service Work
Unlearning Racism in Geoscience (URGE) Leader **(2021, 2023)**

- Organized the participation of the Princeton University Dept. of Geoscience and Atmospheric & Oceanic Sciences in the NSF Funded URGE Program. Managed the 45 member GEO/AOS Pod, including faculty, graduate students, undergraduate students, and staff during the 16-week program.
- Led a 10+ member "podlet" during weekly meetings to discuss live interviews with DEIA leaders and to contribute to biweekly deliverables. Worked with other podlet leaders to combine and summarize 8 deliverables for submission to URGE organizers.
- Co-led an 8-person group to continue URGE efforts following the initial program. Researched the departmental procedure for hiring faculty and reviewing graduate student applications and compared it to similar programs at other universities and recommendations from a literature review. Summarized findings and recommended changes to the procedure for prospective faculty and graduate student interviews.

WORKSHOPS & PROFESSIONAL DEVELOPMENT:

Ocean Data Labs Pedagogy Workshop *NSF OOI & Ocean Data Labs Team,* Cape May, NJ
Hosted by the National Science Foundation's Ocean Observatory Initiative (OOI) **(May 2025)**
Goal: Improve the accessibility and effectiveness of using real-world oceanographic data in undergraduate classrooms using NSF OOI datasets. Participants learned methods for supporting students' development of data literacy skills, reviewed and pilot tested new lab activities, and presented relevant hands-on activities.

Earth Educators' Rendezvous Conference Workshops *NAGT,* Temple University
Hosted by the National Association of Geoscience Teachers (NAGT) **(July 2024)**
Teaching with Investigation and Design in the Undergraduate Science Classroom (TIDeS)
Using Online Quizzes to Promote Effective Student Learning Strategies
Environmental Ethics: Theory and Practice
LGBTQ+ Representation in the Geosciences: Showing Our Students and Staff That They Belong

Teaching Transcript Pedagogy Workshops *McGraw Center*, Princeton University
Trauma-Informed Teaching Practices **(September 2025)**
Developing Inclusive Assessment **(March 2025)**
Designing Effective Syllabi **(November 2024)**
Implementing Active Learning Strategies in the Classroom **(October 2024)**
Making Sense of Student Evaluations **(April 2024)**
Preparing to Write a Meaningful Statement of Teaching Philosophy **(February 2024)**

PUBLICATIONS:

Tang, W., Fortin, S. G., Intrator, N., **Lee, J. A.**, Kunes, M. A., Jayakumar, A., & Ward, B. B. (2025). Contrasting Oxygen Sensitivities of Ammonia Oxidation and Nitrous Oxide Production in Estuarine Waters. Environmental Science & Technology.

Lee, J. A., Vineis, J. H., Poupon, M. A., Resplandy, L., & Ward, B. B. (2025). Phytoplankton community succession and biogeochemistry in a bloom simulation experiment at an estuary-ocean interface. Biogeosciences, 22(18), 4743-4761.

Ombres, E., Benway, H., Bisson, K., Larkin, A., Perotti, E. ..., **Lee, J.,** ... & Zhu, Y. (2025). Tools in Harmony: Integrating Observations and Models for Improved Understanding of a Changing Ocean. Oceanography, 38.

Tang, W., Fortin, S. G., Intrator, N., **Lee, J. A.**, Kunes, M. A., Jayakumar, A., ... & Ward, B. B. (2025) Similar Oxygen Sensitivities of Different Steps of Denitrification in Estuarine Waters. Environmental science & technology.

PUBLICATIONS (CONTINUED):

Tang, W., Da, F., Tracey, J. C., Intrator, N., Kunes, M. A., **Lee, J. A.**, ... & Ward, B. B. (2024). Nutrient management offsets the effect of deoxygenation and warming on nitrous oxide emissions in a large US estuary. Science Advances, 10(51), eadq5014.

Fagan, A. J., Tanioka, T., Larkin, A. A., **Lee, J. A.**, Garcia, N. S., & Martiny, A. C. (2024). Elemental stoichiometry of particulate organic matter across the Atlantic Ocean. *Biogeosciences*, *21*(19), 4239-4250.

Tang, W., Fortin, S. G., Intrator, N., **Lee, J. A.**, Kunes, M. A., Jayakumar, A., & Ward, B. B. (2024). Determination of site‐specific nitrogen cycle reaction kinetics allows accurate simulation of in situ nitrogen transformation rates in a large North American estuary. *Limnology and Oceanography*, *69*(8), 1757-1768.

Tang, W., Jayakumar, A., Sun, X., Tracey, J. C., Carroll, J., Wallace, E., **Lee, J. A.**, Nathan, L., & Ward, B. B. (2022). Nitrous oxide consumption in oxygenated and anoxic estuarine waters. *Geophysical Research Letters*, *49*(21), e2022GL100657.

Tanioka, T., Larkin, A.A., Moreno, A.R., Brock, M.L., Fagan, A.J., Garcia, C.A., Garcia, N.S., Gerace, S.D., **Lee, J.A.**, Lomas, M.W. and Martiny, A.C. (2022). Global ocean particulate organic phosphorus, carbon, oxygen for respiration, and Nitrogen (GO-POPCORN). *Scientific data*, *9*(1), 688.

Moreno, A. R., Larkin, A. A., **Lee, J. A.**, Gerace, S. D., Tarran, G. A., & Martiny, A. C. (2022). Regulation of the respiration quotient across ocean basins. *AGU Advances*, *3*(5), e2022AV000679.

Lee, J. A., Garcia, C. A., Larkin, A. A., Carter, B. R., & Martiny, A. C. (2021). Linking a latitudinal gradient in ocean hydrography and elemental stoichiometry in the eastern Pacific Ocean. *Global Biogeochemical Cycles*, *35*(5), e2020GB006622.

Ustick, L. J., Larkin, A. A., Garcia, C. A., Garcia, N. S., Brock, M. L., **Lee, J. A.**, ... & Martiny, A. C. (2021). Metagenomic analysis reveals global-scale patterns of ocean nutrient limitation. *Science*, *372*(6539), 287-291.

Larkin, A. A., Garcia, C. A., Garcia, N., Brock, M. L., **Lee, J. A.**, Ustick, L. J., ... & Martiny, A. C. (2021). High spatial resolution global ocean metagenomes from Bio-GO-SHIP repeat hydrography transects. *Scientific data*, *8*(1), 107.

Moreno, A. R., Garcia, C. A., Larkin, A. A., **Lee, J. A.**, Wang, W. L., Moore, J. K., ... & Martiny, A. C. (2020). Latitudinal gradient in the respiration quotient and the implications for ocean oxygen availability. *Proceedings of the National Academy of Sciences*, *117*(37), 22866-22872.

MANUSCRIPTS IN PROGRESS:

Lee, J. A. & Ward, B. B. Drivers of temporal co-occurrence patterns and microeukaryote community dynamics in a multispecies diatom bloom. Submitted.

Lee, J. A. & Ward, B. B. Cross domain co-occurrence network analysis of biotic associations and community resilience of a Chesapeake Bay diatom bloom. *In prep*.

CONFERENCES & SYMPOSIUMS:

Lee, J. A. & Ward, B. B. (2026). Drivers of temporal co-occurrence patterns and microeukaryote community dynamics in a multispecies diatom bloom [***Scientific Talk***]. In *Chesapeake Community Research Symposium 2026.* Crowne Plaza Hotel, Annapolis, MD, United States.

Lee, J. A., Vineis, J. H., & Ward, B. B. (2024). Phytoplankton community succession and associated biological productivity measurements in a mesocosm bloom simulation at the Chesapeake Bay estuary-ocean interface [***Poster Presentation***]. In *Ocean Sciences Meeting 2024.* New Orleans Convention Center, New Orleans, LA, United States.

Lee, J. A., Garcia, C. A., & Martiny, A. C. (2019). Latitudinal variation of particulate organic matter stoichiometry in the Pacific Ocean [***Poster Presentation***]. In the *Southern California Geobiology Symposium.* CalTech, Pasadena, CA, United States.

TECHNICAL & ANALYTICAL SKILLS:

Laboratory and Field Techniques
Elemental Analyzer Isotope Ratio Mass Spectrometry (EA-IRMS)
Spectrophotometry & fluorometry
Metabarcoding analysis
    Including DNA extraction, PCR amplification, & Qubit/fluorometric quantification
Micro- and mesocosm seawater incubations
    Including gas manipulation and isotope tracer experiments

Computer Software

Data analysis & visualisation:	Proficient in R, MATLAB, & Cytoscape
	Familiar with Python, ArcGIS, & QIIME2
Education software:	Canvas, Blackboard, Gradescope, & Zoom
Document typesetting:	LaTeX & simple Markdown software
General use:	Microsoft Office Suite & Google Suite

Teaching
Classroom management
Assessment development
Syllabus design
Small group and 1-on-1 instruction
Communication skills
Organizational skills
Problem solving and adaptability
Leadership and collaboration

AWARDS & FUNDING:

Grants and Fellowships

2025	*Program for Community College Engagement*, Princeton University Community College Teaching Fellowship (CCTF)	
2020-2025	Assistantship in Instruction and Research	Princeton University
2019	First Year Fellowship in Natural Science and Engineering	Princeton University
2018	Undergraduate Research Opportunities Program (UROP)	UC Irvine

Awards

2024	Departmental Teaching Award (Dept. of Geosciences)	Princeton University
2021	Service and Outreach Award (Dept. of Geosciences)	Princeton University
2018	Outstanding Senior (Dept. of Earth System Science)	UC Irvine